

Problem Sheet 3

Exercise 1

Consider a set of independent and identically distributed observations $X = \{x_1, x_2, \dots, x_N\}$ drawn from a normal distribution with unknown mean μ and standard deviation σ , denoted by

$$X_i \sim N(\mu, \sigma^2)$$

Given that the Maximum Likelihood Estimate (MLE) for the mean μ has already been derived as:

$$\hat{\mu} \sim \frac{1}{N} \sum_{i=1}^N x_i$$

derive the MLE for the standard deviation σ (denoted as $\hat{\sigma}$), in terms of $\hat{\mu}$ and the observed data.

Solution:

Proof. Since

$$X_i \sim N(\mu, \sigma^2)$$

and x_i i.i.d, so the likelihood function can be written as:

$$L(\mu, \sigma) = \prod_{i=1}^N f(x_i | \mu, \sigma) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x_i - \mu)^2 / 2\sigma^2)$$

Here, to simplify calculation, we can use log-likelihood:

$$\ell(\mu, \sigma) = \log L(\mu, \sigma) = \sum_{i=1}^N [-\log(\sqrt{2\pi}\sigma) - (x_i - \mu)^2 / 2\sigma^2] = -\frac{N}{2} \log(2\pi) - N \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2$$

So we can compute the derivative of σ :

$$\frac{\partial \ell}{\partial \sigma} = -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^N (x_i - \mu)^2$$

and we let derivative equal to 0, we can get:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Then we substitute $\mu = \hat{\mu}$:

$$\hat{\sigma} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2}$$

□

Exercise 2

Consider the Bernoulli process $y_i \in \{0, 1\}, i \in \mathbb{N}$ with

$$p(y_i = 1 \mid x_i) = \frac{1}{1 + \exp(\beta(\theta - x_i))}$$

with parameters β and θ . The log-likelihood function for this model is given by

$$\log p(Y \mid X) = \sum_{i=1}^N [-y_i(\beta(\theta - x_i)) - \log(1 + \exp(-\beta(\theta - x_i)))]$$

where $Y = \{y_1, \dots, y_n\}$ and $X = \{x_1, \dots, x_n\}$.

1. Compute the gradients of the log-likelihood function with respect to the parameters.
2. Write down the update equation of the gradient optimization procedure for the parameters. Is this a case of gradient ascent or descent?
3. Put your algorithm to the test. For this, translate your algorithm into code, simulate ground truth data where you are free to choose n , β , and θ , and validate whether you can recover the ground truth parameters. Explain your validation approach.

Solution:

Proof. 1. Let $z_i = \beta(\theta - x_i)$, then

$$p(y_i = 1 \mid x_i) = \frac{1}{1 + \exp(z_i)}$$

So the log-likelihood function for this model is

$$\ell(\beta, \theta) = \log p(Y \mid X) = \sum_{i=1}^N [-y_i z_i - \log(1 + \exp(-z_i))]$$

And for z_i , it is easy to compute that:

$$\frac{\partial \ell_i}{\partial z_i} = -y_i + \frac{1}{1 + \exp(z_i)} = -y_i + p_i$$

So the gradients of the log-likelihood function are:

$$\frac{\partial \ell}{\partial \beta} = \frac{\partial \ell}{\partial z} \frac{\partial z}{\partial \beta} = \sum_{i=1}^N (-y_i + p_i) \frac{\partial z_i}{\partial \beta} = \sum_{i=1}^N (-y_i + p_i)(\theta - x_i) \quad (1)$$

$$\frac{\partial \ell}{\partial \theta} = \frac{\partial \ell}{\partial z} \frac{\partial z}{\partial \theta} = \sum_{i=1}^N (-y_i + p_i) \frac{\partial z_i}{\partial \theta} = \sum_{i=1}^N (-y_i + p_i)\beta \quad (2)$$

2. From the above results, we can easily get the update equation of the gradient optimization procedure:

$$\beta^{t+1} = \beta^t + \eta^t \sum_{i=1}^N (-y_i + p_i^t)(\theta^t - x_i) \quad (3)$$

$$\theta^{t+1} = \theta^t + \eta^t \beta^t \sum_{i=1}^N (-y_i + p_i^t) \quad (4)$$

This is a case of gradient ascent, since what we do is maximise the log-likelihood.

3. See below.

□

Ex2.3

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In [1]: import numpy as np
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In [2]: def probability_of_z(z):
        return 1 / (1 + np.exp(z))

def log_likelihood(X, Y, beta, theta):
    z = beta * (theta - X)
    return np.sum(- Y * z - np.log(1 + np.exp(- z)))

def gradient(X, Y, beta_init, theta_init, lr=0.01, n_iters=1000, print_ev
    beta = beta_init
    theta = theta_init
    history = []

    for _ in range(n_iters):
        z = beta * (theta - X)
        p = probability_of_z(z)

        grad_beta = np.sum((p - Y) * (theta - X))
        grad_theta = np.sum((p - Y) * beta)

        beta += lr * grad_beta
        theta += lr * grad_theta

        if _ % print_every == 0 or _ == n_iters - 1:
            ll = log_likelihood(X, Y, beta, theta)
            history.append(ll)
            print(f"Iteration {_}: beta = {beta:.4f}, theta = {theta:.4f}")

    return beta, theta, history

# ground truth generation
def simulate_data(n=200, beta_true=2.0, theta_true=1.0, x_low=-3.0, x_high
    rng = np.random.default_rng(seed)
    X = rng.uniform(x_low, x_high, size=n)
    z = beta_true * (theta_true - X)
    p = 1.0 / (1.0 + np.exp(z))
    Y = rng.binomial(1, p)
    return X, Y
```

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In [3]: # fit the model
        beta_true = 3.0
        theta_true = -2.0

        beta_init = 0.1
        theta_init = 0.1
        lr = 3e-4
        n_iters = 2000
        print_every = 100

        X, Y = simulate_data(n=200, beta_true=beta_true, theta_true=theta_true, s
```

```
estimated_beta, estimated_theta, history = gradient(X, Y, beta_init, thet
```

```
Iteration 0: beta = 0.1259, theta = 0.0979, Log-Likelihood = -126.8045
Iteration 100: beta = 0.9401, theta = -0.8447, Log-Likelihood = -67.5983
Iteration 200: beta = 1.4055, theta = -1.6466, Log-Likelihood = -38.2384
Iteration 300: beta = 1.7269, theta = -1.9682, Log-Likelihood = -30.9138
Iteration 400: beta = 1.9452, theta = -2.0412, Log-Likelihood = -29.0877
Iteration 500: beta = 2.1150, theta = -2.0416, Log-Likelihood = -28.1230
Iteration 600: beta = 2.2577, theta = -2.0303, Log-Likelihood = -27.4385
Iteration 700: beta = 2.3817, theta = -2.0206, Log-Likelihood = -26.9224
Iteration 800: beta = 2.4913, theta = -2.0135, Log-Likelihood = -26.5206
Iteration 900: beta = 2.5891, theta = -2.0084, Log-Likelihood = -26.2002
Iteration 1000: beta = 2.6775, theta = -2.0046, Log-Likelihood = -25.9397
Iteration 1100: beta = 2.7577, theta = -2.0016, Log-Likelihood = -25.7245
Iteration 1200: beta = 2.8312, theta = -1.9993, Log-Likelihood = -25.5445
Iteration 1300: beta = 2.8987, theta = -1.9974, Log-Likelihood = -25.3923
Iteration 1400: beta = 2.9611, theta = -1.9959, Log-Likelihood = -25.2624
Iteration 1500: beta = 3.0190, theta = -1.9946, Log-Likelihood = -25.1507
Iteration 1600: beta = 3.0729, theta = -1.9935, Log-Likelihood = -25.0539
Iteration 1700: beta = 3.1232, theta = -1.9926, Log-Likelihood = -24.9696
Iteration 1800: beta = 3.1702, theta = -1.9918, Log-Likelihood = -24.8957
Iteration 1900: beta = 3.2144, theta = -1.9912, Log-Likelihood = -24.8306
Iteration 1999: beta = 3.2555, theta = -1.9906, Log-Likelihood = -24.7737
```

Exercise 3

1. Posterior distribution of μ

Since X is assumed to be log-normal, the model can be written on the log-scale as

$$Y_i = \ln(X_i), \quad Y_i \sim \mathcal{N}(\mu, \sigma^2).$$

Here, μ is the unknown parameter of interest. Following the Bayesian idea, the posterior is proportional to likelihood times prior:

$$p(\mu \mid Y) \propto p(Y \mid \mu)p(\mu).$$

For a log-normal distribution,

$$\sigma^2 = \ln \left(1 + \frac{\text{Var}(X)}{\mathbb{E}(X)^2} \right),$$

and

$$\mu = \ln(\mathbb{E}(X)) - \frac{\sigma^2}{2}.$$

For the observed data,

$$\mathbb{E}(X) = 250, \quad \text{Var}(X) = 100.$$

Therefore,

$$\sigma^2 = \ln \left(1 + \frac{100}{250^2} \right),$$

and the observed estimate for the log-mean is

$$\hat{\mu} = \ln(250) - \frac{1}{2} \ln \left(1 + \frac{100}{250^2} \right).$$

For the prior information,

$$\mathbb{E}_{\text{prior}}(X) = 300, \quad \text{Var}_{\text{prior}}(X) = 150.$$

Thus,

$$\tau^2 = \ln \left(1 + \frac{150}{300^2} \right),$$

and

$$\mu_{\text{prior}} = \ln(300) - \frac{1}{2} \ln \left(1 + \frac{150}{300^2} \right).$$

The prior distribution is therefore

$$\mu \sim \mathcal{N}(\mu_{\text{prior}}, \tau^2).$$

The likelihood on the log-scale is normal. Using the sample mean on the log-scale as the estimate,

$$\hat{\mu} \mid \mu \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Since both likelihood and prior are normal, this is a conjugate Bayesian update. Therefore, the posterior is also normal:

$$\mu \mid Y \sim \mathcal{N}(\mu_{\text{post}}, \sigma_{\text{post}}^2).$$

The posterior variance is

$$\sigma_{\text{post}}^2 = \left(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\right)^{-1}.$$

The posterior mean is

$$\mu_{\text{post}} = \sigma_{\text{post}}^2 \left(\frac{\mu_{\text{prior}}}{\tau^2} + \frac{n\hat{\mu}}{\sigma^2}\right).$$

Substituting the values gives

$$\sigma_{\text{post}}^2 = \left(\frac{1}{\ln\left(1 + \frac{150}{300^2}\right)} + \frac{30}{\ln\left(1 + \frac{100}{250^2}\right)}\right)^{-1}.$$

Also,

$$\mu_{\text{post}} = \sigma_{\text{post}}^2 \left(\frac{\ln(300) - \frac{1}{2} \ln\left(1 + \frac{150}{300^2}\right)}{\ln\left(1 + \frac{150}{300^2}\right)} + \frac{30 \left[\ln(250) - \frac{1}{2} \ln\left(1 + \frac{100}{250^2}\right)\right]}{\ln\left(1 + \frac{100}{250^2}\right)}\right).$$

Numerically,

$$\sigma_{\text{post}}^2 \approx 0.0000516,$$

and

$$\mu_{\text{post}} \approx 5.52631.$$

Thus,

$$\boxed{\mu \mid Y \sim \mathcal{N}(5.52631, 0.0000516)}$$

2. Posterior mean and variance

From the posterior distribution,

$$\boxed{\mu_{\text{post}} \approx 5.52631}$$

and

$$\boxed{\sigma_{\text{post}}^2 \approx 0.0000516.}$$

The posterior mean is closer to the observed data than to the prior mean. This is reasonable because the posterior combines the prior information with the data from $n = 30$ participants.

3. Predictive distribution

For a new participant,

$$Y_{\text{new}} = \ln(X_{\text{new}}).$$

The predictive distribution also includes the uncertainty about μ :

$$Y_{\text{new}} \mid Y \sim \mathcal{N}(\mu_{\text{post}}, \sigma^2 + \sigma_{\text{post}}^2).$$

Therefore,

$$X_{\text{new}} \mid Y \sim \text{LogNormal}(\mu_{\text{post}}, \sigma^2 + \sigma_{\text{post}}^2).$$

Using the values,

$$\sigma^2 + \sigma_{\text{post}}^2 = \ln\left(1 + \frac{100}{250^2}\right) + \sigma_{\text{post}}^2.$$

Numerically,

$$\sigma^2 + \sigma_{\text{post}}^2 \approx 0.0016506.$$

So,

$$X_{\text{new}} \mid Y \sim \text{LogNormal}(5.52631, 0.0016506)$$

The predictive mean is

$$\mathbb{E}[X_{\text{new}} \mid Y] = \exp\left(\mu_{\text{post}} + \frac{\sigma^2 + \sigma_{\text{post}}^2}{2}\right).$$

Numerically,

$$\mathbb{E}[X_{\text{new}} \mid Y] \approx 251.4 \text{ ms.}$$

4. Interpretation

The prior expected reaction time was about 300 ms. After updating the prior with the observed data, the posterior predictive mean is about 251.4 ms.

Since a smaller reaction time means a faster response, the result suggests that the training was beneficial. The expected reaction time after training is lower than the prior expectation.

Exercise 4 (Hypothesis Testing)

Task 1: Formulate the null and alternative hypotheses

Hypotheses:

$$H_0 : \mu \leq 50$$

$$H_1 : \mu > 50$$

Task 2: Conduct a one-sample z-test, calculate the test statistic, and determine the p-value.

The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{54 - 50}{10/\sqrt{25}} = \frac{4}{2} = 2$$

The p-value is

$$p = P(Z > 2)$$

From the standard normal table,

$$P(Z \leq 2) = 0.9772$$

Therefore,

$$p = 1 - 0.9772 = 0.0228$$

Task 3: Based on the results, conclude whether to reject or fail to reject the null hypothesis, and interpret the findings in the context of cognitive enhancement.

Since $p = 0.0228 < 0.05$, we reject the null hypothesis. So there is sufficient evidence to conclude that the mnemonic training program significantly increases recall scores.

Exercise 5 ((Bootstrapping))

Task 1:

We set $\beta_1 = 0$, the model becomes

$$P(y_i = 1) = \frac{1}{1 + \exp(-\beta_0)}$$

Since a proportion of $\frac{1}{3}$ of the responses are $y_i = 1$,

$$P(y_i = 1) = \frac{1}{3}$$

So

$$\frac{1}{1 + \exp(-\beta_0)} = \frac{1}{3}$$

$$1 + \exp(-\beta_0) = 3$$

$$\exp(-\beta_0) = 2$$

$$-\beta_0 = \ln 2$$

Therefore, an intuitively reasonable choice for β_0 is

$$\beta_0 = -\ln 2$$

Task 2:

We would like to know whether an increase in x_i is associated with a higher response probability.

When $\beta_1 > 0$, increasing x_i makes

$$-\beta_0 - \beta_1 x_i$$

smaller. Therefore,

$$\exp(-\beta_0 - \beta_1 x_i)$$

becomes smaller, which decreases the denominator

$$1 + \exp(-\beta_0 - \beta_1 x_i).$$

So the probability $P(y_i = 1 \mid x_i)$ increases.

Therefore, the hypotheses are

$$H_0 : \beta_1 \leq 0$$

$$H_1 : \beta_1 > 0$$

Task 3:

We use the estimated coefficient as the test statistic:

$$T_{\text{obs}} = \hat{\beta}_1.$$

The bootstrap procedure is:

- To approximate the null distribution, assume the null hypothesis is true.
- Under the null model,

$$P(y_i = 1) = \frac{1}{1 + \exp(-\beta_0)}.$$

Using the proportion,

$$\hat{p}_0 = \frac{1}{3}.$$

- For $b = 1, \dots, B$, generate bootstrap responses

$$y_i^{*(b)} \in \{0, 1\}$$

- For each bootstrap sample, fit the model and compute

$$T^{*(b)} = \hat{\beta}_1^{*(b)}.$$

- The p-value is

$$p \approx \frac{1}{B} \sum_{b=1}^B I(T^{*(b)} \geq T_{\text{obs}}).$$

- If the p-value is smaller than the significance level, reject H_0 .